

Software Test Plan (STP)

Online Internship & Placement Preparation Portal

Version: 1.0

Team No: 14

Course: Software Engineering

Date: 16-09-2026

Team Members

SRN	Name	Contribution
PES2UG24CS043	Akhila T	Test plan structure, coordination, this document
PES2UG24CS022	Abhinav Ranjith Das	Functional test design (Auth, Drives)
PES2UG24CS023	Abhirama H Y	Security & non-functional testing
PES2UG24CS039	Aishwarya	Traceability, review and formatting

Revision History

Version	Date	Author	Change summary	Approval
1.0	16-09-2026	Team 14	Initial draft of the Software Test Plan for the Online Internship & Placement Preparation Portal	

Approvals

Role	Name	Signature / Email	Date
Course Coordinator			

Table of Contents

1. Introduction
2. Test Items
3. Features to be Tested
4. Features Not to be Tested
5. Test Approach / Strategy
6. Test Environment
7. Test Schedule
8. Test Deliverables
9. Roles and Responsibilities
10. Risks and Mitigation
11. Assumptions & Dependencies
12. Suspension & Resumption Criteria
13. Test Case Management & Traceability
14. Test Metrics & Reporting

1. Introduction

1.1 Purpose

This document defines the test plan for Version 1.0 of the Online Internship & Placement Preparation Portal (OIPP). It outlines the objectives, scope, strategy, resources, schedule and responsibilities for testing the system before it is handed over for evaluation.

1.2 Scope

Testing covers the portal's core modules: registration and authentication, profile and resume management, placement/internship drives, mock tests, interview preparation, notifications, and the TPO dashboard/reporting. Excluded from scope: the institution's ERP/HR systems, salary or offer-letter processing, and any external job-posting API integration, since none of these are built in v1.0.

1.3 References

Software Requirements Specification (SRS) v1.0, Software Architecture and Design Specification (SAD) v1.0, both for this project.

1.4 Definitions

TPO (Training & Placement Officer), OTP (One-Time Password), JWT (JSON Web Token), API (Application Programming Interface), RBAC (Role-Based Access Control), CGPA (Cumulative Grade Point Average), SRS (Software Requirements Specification), RTM (Requirements Traceability Matrix).

2. Test Items

- Auth Service (registration, OTP verification, login, password reset)
- Profile Service (profile creation/edit, resume upload and download)
- Drive Service (posting drives, eligibility filtering, applying, duplicate-application prevention)
- Test Engine Service (aptitude/technical mock tests, timer, auto-grading, history)
- Interview Prep Service (question bank search, mock-interview booking, feedback)
- Notification Service (email/in-app notifications)
- Reporting Service (TPO dashboard statistics, CSV export)

3. Features to be Tested

Features mapped to SRS requirement IDs:

- OIPP-F-001 / F-002 / F-003 — Registration, login, password reset
- OIPP-F-004 / F-005 / F-006 — Profile creation and resume upload/template
- OIPP-F-007 / F-008 / F-009 / F-010 — Posting a drive, eligibility filtering, applying, duplicate prevention
- OIPP-F-011 / F-012 / F-013 — Mock tests, auto-grading, attempt history
- OIPP-F-014 / F-015 / F-016 — Interview question bank, mock-interview booking, feedback
- OIPP-F-017 / F-018 — Application status tracking and notifications
- OIPP-F-019 / F-020 — TPO dashboard statistics and CSV export
- OIPP-NF-001 / NF-005 — Response time and concurrent-user load
- OIPP-SR-001 to SR-005 — HTTPS/TLS, password hashing, email-domain restriction, account logout, role-based access control

4. Features Not to be Tested

- Processing of salaries or offer letters after a student is hired (out of scope per SRS)
- Integration with the institution's ERP for auto-filling academic records (future enhancement, not built in v1.0)
- The optional external job-posting API integration (kept as a future enhancement, not built in v1.0)
- Internal working of the third-party email/SMS gateway itself (treated as a trusted external service; only the portal's use of it is tested)

5. Test Approach / Strategy

5.1 Levels

- Unit tests — individual functions/modules (e.g. eligibility check, OTP generation)
- Integration tests — service-to-service and service-to-database interaction (e.g. Drive Service ↔ Database ↔ Notification Service)
- System tests — end-to-end flows through the UI (e.g. register → build profile → apply to a drive)
- Acceptance tests — course-evaluator / stakeholder walkthrough against the SRS acceptance criteria

5.2 Types

- Functional testing — all features in Section 3
- Regression testing — re-run of prior passing cases after each new feature or bug fix
- Performance testing — response time (OIPP-NF-001) and concurrent-user load (OIPP-NF-005)
- Security testing — see Section 5.4
- Usability / accessibility testing — screen-reader and contrast checks (OIPP-NF-004)

5.3 Entry and Exit Criteria

Entry Criteria: a stable build is available, test data is prepared, and the test environment is set up.

Exit Criteria: all high-priority functional requirements are implemented and pass their acceptance test, no critical NFR fails during testing, and the RTM (Section 13) shows every listed test case executed with a Pass status.

5.4 Security Validation

- Verify passwords are stored as salted bcrypt hashes, never in plain text (OIPP-SR-002)
- Verify all traffic is served over HTTPS/TLS 1.2 or higher, with non-HTTPS requests rejected (OIPP-SR-001)
- Verify registration is rejected for a non-institution email address (OIPP-SR-003)
- Verify an account is locked for 15 minutes after 5 consecutive failed login attempts (OIPP-SR-004)
- Verify role-based access control: a student account is rejected with a 403 when attempting TPO-only actions (OIPP-SR-005)

6. Test Environment

Hardware: a laptop, desktop or phone with a browser; a webcam/microphone only for testing the optional mock-interview video feature.

Software: React front end, Node.js/Express backend, PostgreSQL database, cloud object storage — deployed to a staging instance that mirrors the production stack described in the SAD.

Tools: Postman (API testing), a free/open-source load-testing tool such as Apache JMeter (performance testing), browser developer tools and a screen reader (accessibility checks), GitHub Issues (defect tracking).

Test Data: dummy student accounts covering different branches, CGPA values and backlog counts; sample drives with varying eligibility criteria; sample PDF resumes; dummy interview questions.

7. Test Schedule

Milestone	Date
Test case design	22-Sep-2026
Test environment setup	24-Sep-2026
Test execution start	25-Sep-2026
Test execution end	08-Oct-2026
Acceptance walkthrough / demo	10-Oct-2026

8. Test Deliverables

- Test Plan (this document)
- Test cases (manual)
- Test data
- Test execution logs
- Defect reports
- Test summary report
- Updated Requirements Traceability Matrix (RTM)

9. Roles and Responsibilities

Role	Name	Responsibility
Test Lead	Akhila T	Prepare the test plan, coordinate execution, consolidate the summary report
Test Engineer	Abhinav Ranjith Das	Design and execute functional test cases for Auth and Drive modules, log defects
Test Engineer	Abhirama H Y	Design and execute security and non-functional test cases, log defects
Test Engineer	Aishwarya	Maintain the RTM, review test cases, format and consolidate reports

10. Risks and Mitigation

Risk	Mitigation
Limited team availability due to overlapping coursework/exams	Build buffer time into the schedule; split test cases evenly among team members
No budget for paid performance-testing tools	Use free/open-source tools such as Apache JMeter
Test/staging environment differs from eventual production hosting	Keep the same technology stack (Node.js, PostgreSQL) locally as described in the SAD
OTP/notification testing depends on an external email service	Use a test/sandbox SMTP account during testing rather than the live one
A module is not ready by the planned test-execution start date	Prioritise smoke testing of completed modules first; adjust schedule for the rest

11. Assumptions & Dependencies

- A reasonably stable build of all core modules is available before system testing begins
- Test data (dummy student accounts, drives, resumes) is prepared in advance by the team
- The staging environment has network access to a sandbox email/SMTP service for OTP and notification testing
- All four team members are available through the test-execution window in Section 7

12. Suspension & Resumption Criteria

Suspend testing if:

- The test environment is unavailable for more than 4 hours
- The build is too unstable to proceed, blocking more than 30% of planned test cases

Resume testing if:

- Blocking defects are resolved
- The environment is stabilised

13. Test Case Management & Traceability

The RTM ensures every SRS requirement maps to at least one test case. A sample is shown below; the full matrix is maintained alongside the SRS and updated as execution progresses. Status: N = Not tested, P = Passed, F = Failed, A = Automated.

Req ID	Requirement (short)	Test case(s)	Status	Comments
OIPP-F-001	Register & verify via OTP	TC-Reg-01	N	
OIPP-F-002	Login	TC-Auth-01	N	
OIPP-F-008	Filter drives by eligibility	TC-Drv-02	N	
OIPP-F-009	Apply to a drive	TC-Drv-03	N	
OIPP-F-011	Take a mock test	TC-Test-01	N	
OIPP-NF-001	Response time target	TC-Perf-01	N	
OIPP-SR-002	Password hashing	TC-Sec-01	N	
OIPP-SR-005	Role-based access control	TC-Sec-05	N	

14. Test Metrics & Reporting

Metrics collected:

- % of test cases executed
- % passed / failed
- Defect count and severity
- Requirement coverage (from the RTM)

Reports:

- Test execution status, updated as testing progresses
- Final Test Summary Report